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To cite this article: Xinhui Kang, Huiqi Wan & Shuyao Li (14 Aug 2025): Combining Kansei Engineering and SSA-XGBoost for new energy vehicle front-face form design, Journal of Engineering Design, DOI: [10.1080/09544828.2025.2546838](https://doi.org/10.1080/09544828.2025.2546838)

To link to this article: <https://doi.org/10.1080/09544828.2025.2546838>



Published online: 14 Aug 2025.



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Combining Kansei Engineering and SSA-XGBoost for new energy vehicle front-face form design

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ABSTRACT

With the worsening global energy crisis and environmental pollution, new energy vehicles (NEV) have emerged as a vital solution for sustainable development. As a vehicle's most visually striking part, the front-face significantly influences user perception and market competitiveness. This paper proposes a novel front-face design method for NEV by integrating Kansei Engineering (KE) with the Sparrow Search Algorithm-Extreme Gradient Boosting (SSA-XGBoost) to enhance emotional appeal. The research involves four stages. First, representative Kansei words are extracted using the ChatGPT language model and prioritised through the Entropy-AHP method. Second, to break free from existing market design limitations, the Stable Diffusion Model (SDM) generates 99 new NEV images, from which a morphological database and deconstruction table are created using morphological analysis. Third, the SSA-XGBoost model builds a mapping between user emotions and design features, identifying optimal parameter combinations. Finally, use Rhino to construct the three-dimensional model and conduct a sensory and rational evaluation of the car's frontal shape design through aerodynamic performance computational fluid dynamics (CFD) simulations and expert evaluations. Experimental results show that this method achieves higher prediction accuracy and customer satisfaction compared to traditional machine learning approaches, providing a creative and efficient framework for NEV front-face design.

ARTICLE HISTORY

Received 11 April 2025
Accepted 8 August 2025

KEYWORDS

Kansei Engineering;
SSA-XGBoost; NEV; form
design; CFD

1. Introduction

In the context of the global automotive industry's transformation towards low-carbon and environmental protection, the NEV market is experiencing unprecedented vigorous development. NEV has become a top priority for the future development of the automotive industry (Y. Wang et al. 2025). In 2024, the global sales volume of NEV exceeded 17 million units, representing a year-on-year growth of approximately 25%. NEV not only reduces the consumption of fossil energy but also decreases the emissions of atmospheric pollutants. With the continuous advancement of technology, significant breakthroughs have been achieved in aspects such as the driving range and charging facilities of NEV. The functional differences between products are gradually narrowing, leading to consumers' demands for

NEV no longer being limited to essential transportation functions but progressively shifting towards the customer's emotional experience (Wang, Sinha et al., 2024; Zheng et al. 2022). Therefore, the visual aesthetics of automotive styling has become a key consideration in the design process, which can significantly increase the vehicle's market share (Avikal, Singh, and Rashmi 2020). An emotionally rich form design of NEV can not only attract consumers' attention visually but also enhance the market competitiveness and user loyalty of NEV. The front-face of a vehicle is usually the focal point of automotive design; it not only conveys abundant emotional information but also can arouse customers' purchasing interest and emotional resonance (Sun and Park 2024). In the design of the vehicle's front-face, classic form features enable manufacturers to stand out in the highly competitive market and help consumers establish a deep emotional connection with the brand. For example, the tiger-nose front-face of Kia vehicles, the kidney grille of BMW, and the hawk-eye headlights of Audi, etc. Therefore, this study proposes an innovative design method for the front-face of NEV.

The traditional NEV front design process is to investigate consumer needs through the market, then create conceptual sketches based on designers' subjective experience, and finally convert these sketches into three-dimensional models (Jing et al. 2021). With the arrival of the experience economy, the market's expectation for NEV has long gone beyond simple, functional satisfaction. It pays more attention to the emotional experience and psychological satisfaction brought by the car. The explosion of China's new power car brand, Xiaomi Automobile, has confirmed this, with more than 50,000 units of pre-orders within the week of its launch and subsequent deliveries continuing to grow. The shape design combined with the sliding back line and the through-light belt of the Xiaomi car is not only fashionable but also can reduce wind resistance and meet the aesthetic needs of users.

KE is a method of transforming customer emotions into product design features. In the field of automotive design, it mainly provides automotive design guidance by associating users' emotional images of automotive products (H. Zhao, Sharudin, and Lv 2024; Qi and Kim 2024). Unlike traditional qualitative emotional design methods, KE is a quantitative product design technique guided by consumer needs, which can transform consumers' vague Kansei into specific design elements (Kang, You, and Luo 2025; Lévy 2013). The first proposal of KE in the automotive field can be traced back to the mid-1980s, after which many scholars introduced KE into the field of automotive design. Zang et al. (2024) applied KE to automotive styling design. Yang, Cheng, and Wang (2019) proposed an innovative car seat design method using KE. KE reduces the deviation in the understanding of products between designers and consumers by exploring the correlation between people's emotional experiences and product design elements (Tang et al. 2024). KE can assist designers in gaining insights into consumers' Kansei cognition regarding aspects such as the exterior and interior of automobiles. At present, KE has two core characteristics: (1) accurately capturing consumers' Kansei needs and (2) Establishing a mapping model between customer Kansei requirements and product design features and transforming it into specific product design specifications (Wang, Niu et al., 2024).

Traditional KE has some defects in processing the ordering of complex Kansei of customers and constructing mapping models (X. Li and Li 2024). On the one hand, when capturing consumers' Kansei needs, KE often relies on questionnaires and group interviews. These methods are highly subjective, and it is difficult for consumers to accurately express their potential needs, resulting in the collected information failing to reflect consumers'

Kansei expectations of products. Standard calculation methods for handling the weights of users' Kansei needs include the Analytic Hierarchy Process (AHP), Kano model and Entropy method, etc. However, there are obvious drawbacks when using these quantitative methods alone. AHP has limited adaptability to complex demand structures, users' subjective factors overly influence the Kano model, and the Entropy method is insufficient in reflecting Kansei. Therefore, this paper proposes the Entropy-AHP method that combines subjective and objective evaluations to handle the ranking of users' emotional needs. It effectively integrates Kansei data and structured analysis, improves problems such as subjective and objective biases in traditional methods and the difficulty in dealing with complex information, enhances the accuracy and consistency of decision-making.

On the other hand, classical KE mapping models include linear analysis methods such as multiple regression analysis and factor analysis. However, these methods are unable to accurately measure customers' emotions that are non-linearly distributed and do not follow a normal distribution. While mapping models in machine learning, such as Convolutional Neural Network (CNN), Genetic Algorithm (GA), and Support Vector Regression (SVR), can effectively address the non-linear characteristics of Kansei intentions, they also have corresponding drawbacks. For example, due to its inherent characteristics, CNN model has a strong dependence on data and poor interpretability, making it difficult to integrate multi-modal Kansei information. The GA is based on a fitness function during the search process, making it difficult to guide the search direction precisely. The SVR has poor adaptability to complex and fuzzy Kansei data in Kansei mapping and has difficulties in handling high-dimensional sparse data (Li, Feng et al., 2022; Wang, Zhang et al., 2024). With the rapid development of Artificial Intelligence (AI) technology, many scholars have proposed using AI technology to effectively map users' Kansei intentions to design elements, thereby enabling a deeper understanding of users' true needs (Kang et al. 2025; Lu et al. 2024; Z. Chen 2024).

Therefore, the XGBoost algorithm proposed in 2016 is introduced into the research of KE in this paper. This AI algorithm is an efficient improvement of the traditional gradient boosting algorithm and has demonstrated powerful performance in the fields of data mining and prediction. It achieves efficient training through parallel computing and distributed operation. When dealing with large-scale high-dimensional data, XGBoost can not only automatically fill in missing values but also enhance its robustness and prevent overfitting by means of regularisation (S. Xie et al. 2025; T. Chen and Guestrin 2016). Applying it to automotive form design can efficiently analyse large Kansei data and accurately predict consumers' preferences for automotive form design, providing strong support for designers' scientific decision-making. However, the parameter adjustment of XGBoost during model optimisation depends on manual operation and is prone to falling into local optimality, resulting in low computational efficiency and long training time.

To further improve the classification and prediction performance of the XGBoost model, this paper introduces the SSA to optimise it. The SSA has the ability to effectively explore the potential areas of the global optimum, which can effectively avoid the problem of local optimality. After the SSA optimises the parameters of XGBoost, it enables the model to be in a favourable position at the beginning of training; it has the advantages of converging to a better solution more quickly, improving the prediction accuracy and reducing the number of iterations. At the same time, the SSA helps XGBoost to balance the bias and variance more accurately, avoiding overfitting and underfitting and enhancing the generalisation

ability and stability of the model on both the training set and the test set. In addition, the robustness of the XGBoost model optimised by the SSA is significantly improved. The optimised parameters are highly stable and the model performs stably when facing different data sets and data noises, being less affected by data fluctuations (X. Zhao et al. 2022; Yue et al. 2023).

In summary, this paper proposes a method for the form design of the front-face of NEV that combines KE and SSA-XGBoost. Firstly, with the help of the large language model Chat-GPT, the customers' Kansei needs are directly generated and the SDM is used to generate a large database of NEV that have not been launched on the market. Secondly, Entropy-AHP is adopted to determine the priority ranking of Kansei words and the SSA-XGBoost model is introduced to establish the mapping relationship between customers' key Kansei and the design features of NEV so as to obtain the optimal front-face design combination. SDM automatically generates corresponding three-view diagrams based on the optimal front face shape combination. Through the scoring given by 15 experts and scholars in the automotive field regarding the matching degree of three aesthetic terms and three front faces of the car, it verifies whether the design outcome can meet people's aesthetic needs. Secondly, a 3D model is established based on the three-view diagrams and CFD experiments are used to conduct physical performance tests on the car body design. This research provides an innovative design method for the front face shape design of NEV, thereby enhancing the market competitiveness of NEV. The main contributions are as follows:

- (1) SSA-XGBoost accurately establishes the mapping relationship between customers' Kansei and design features, providing a novel scientific method for product styling design.
- (2) The SDM automatically generates the matching 3D renderings of vehicles according to the features of a single front view, which greatly improves the design efficiency and shortens the research and development cycle.

The organisational framework of this paper is as follows: Section 2 introduces the relevant literature. Section 3 proposes the research process framework. Section 4 presents a specific case of the NEV's front-face form design. The discusses and the research results are proposed at Section 5, and the Section 6 summarises the study and proposes future research directions.

2. Literature review

2.1. Application of KE in automobiles

KE is a specific engineering technology that transforms users' Kansei images into product design features, and it is dedicated to building a bridge between product design and consumers' emotional needs. Automobile products designed using KE not only possess practical functions but also can meet users' psychological and emotional demands. In the era of the vigorous development of the current automotive industry, KE plays an increasingly important role in the field of automotive design (Lai et al. 2022; Lu et al. 2025). Many scholars have carried out in-depth research on KE, aiming to optimise the automotive design process and improve the design quality. Wang, Shi et al. (2024) combined KE

with the SSA-LSTM-Attention model to enhance the emotional experience and customer satisfaction of the intelligent cockpit of NEV. Kang (2021) combined the rough set theory and SVR to conduct the sustainable form design of hybrid electric vehicles. Vieira et al. (2017) used KE to provide the industry with user-centred design guidelines for in-vehicle rubber keyboards.

Although the above-mentioned studies have shown that KE can accurately transform customers' abstract Kansei needs into specific aspects such as the exterior styling and interior layout of automobiles. However, existing studies often require a large investment of time and economic resources in the data collection and model construction processes. Traditional methods of collecting emotional data mainly rely on questionnaires and focus group discussions, which inevitably have the disadvantages of being highly subjective, time-consuming and labour-intensive. In addition, during the data processing stage, the processes of cleaning and complex Kansei data are cumbersome and inefficient, making it difficult to extract valuable information from a large amount of unstructured data.

Therefore, this paper integrates KE with AI technology and applies it to automotive styling design. AI technology, with its powerful data processing capabilities and efficient algorithm models, can quickly handle massive perceptual data. Through natural language processing technology, non-structured data such as online consumer reviews and social media discussions can be analysed to uncover the hidden perceptual needs. By using AI algorithms, a more accurate mapping model between Kansei needs and automotive styling design elements can be constructed, enhancing the accuracy and reliability of the design. This technological integration is expected to break through the inherent limitations of traditional methods, significantly shorten the cycle of automotive styling design, bring significant improvements and innovative breakthroughs to the efficiency of automotive styling design.

2.2. Application of entropy-AHP in emotion calculation

Entropy is a typical objective weighting method based on diversity, which can objectively process information and improve the consistency of decision-making (R. Zhang et al. 2023). Wang, Cheng et al. (2024) utilised Entropy to optimise the design of tourism mascots. L. Liu et al. (2010) employed Entropy to establish an improved method for water quality assessment. However, as an objective measurement method, Entropy is unable to fully consider users' subjective expressions during the process of evaluating the Kansei needs of automobiles.

To obtain an accurate ranking of customers' Kansei needs, this paper introduces the AHP to consider various influencing factors and their interrelationships comprehensively. AHP realises the combination of qualitative and quantitative analysis. It can not only quantify the subjective judgments of experts on qualitative factors but also calculate and determine the weights of various factors by combining existing quantitative data, ranking the alternatives and providing straightforward decision-making suggestions (Costa, Mamede, and Silva 2023; Feng, Hou, and Lan 2024). X. Xie et al. (2023) used AHP to propose an evaluation method for the green design of kindergarten furniture. M. Liu et al. (2023) applied AHP to establish an evaluation system for restaurant chair design that can meet users' needs from the perspective of user requirements. The practical applications of AHP in the above

different design scenarios provide a powerful reference for its application in the field of automotive styling design. However, using the AHP method alone is likely to result in less objective calculation results (Liu, Zhang, Ji et al., 2024; Yu et al. 2024).

Therefore, this paper uses the integrated Entropy-AHP method to rank consumers' Kansei needs more accurately, making the research results more scientific and practical. The Entropy-AHP method makes up for the deficiencies of a single process. It not only fully considers the objectivity of the data but also takes into account the subjective preferences of decision-makers, making the decision-making results more reasonable (Xiao et al. 2022). H. Chen (2020) used Entropy-AHP to construct a decision-making model for suppliers of new building materials, effectively selecting suitable suppliers. Nyimbili and Erden (2020) used the Entropy-AHP method to subjectively and objectively evaluate the appropriate areas for planning new urban emergency facilities in Istanbul Province. Y. Zhang, Li, and Li (2024) conducted a study on the progressive planning scheme of the power grid based on Entropy-AHP. Currently, Entropy-AHP has achieved remarkable results in multiple fields, but it has not yet covered the field of automotive design. Therefore, this study innovatively introduces Entropy-AHP to determine the emotional weights of NEV, scientifically balancing the subjective and objective weights of customers.

2.3. SSA-XGBoost

XGBoost is an efficient integrated learning algorithm based on the gradient lifting framework. It avoids overfitting by introducing L1 and L2 regularisation terms to control the complexity of the model, uses feature blocking and cache optimisation technology to realise parallel computation to improve training speed and also supports custom loss functions and multiple data type inputs. It has the advantages of flexibility and expansibility (Friedman 2006; Liu, Zhang, Zhang et al., 2024). XGBoost has been widely used in architecture (Mo et al. 2019), meteorology (Li, An et al., 2022), engineering technology (Qiu et al. 2022) and other fields. However, when XGBoost is faced with a large number of complex Kansei evaluations, it may face the problem of hyperparameter tuning, and traditional grid search methods are time-consuming and easy to fall into the local optimal dilemma (Vadhwani and Thakor 2023).

SSA algorithm simulating sparrow foraging behaviour is introduced to solve the above defects existing in XGBoost. SSA has a strong global search capability, avoids falling into local optimality through the 'search-follower' mechanism, only needs to adjust a few parameters and has low computational complexity (Xue and Shen 2020). The SSA-XGBoost hybrid model has significant advantages; SSA can adaptatively search for the optimal XGBoost hyperparameter combination to improve performance and generalisation ability and can better capture the complex nonlinear relationship between user Kansei requirements and NEV positive face design elements (Gao et al. 2022). SSA-XGBoost integrates meta-heuristic optimisation and ensemble learning and makes remarkable progress in solving the XGBoost parameter-sensitive Kansei problem and improving prediction accuracy. Xu et al. (2023) proposed to use of SSA-XGBoost to predict the problem of mine-induced ground lift. This paper proposes SSA-XGBoost to deeply explore the optimisation design of the front face shape of NEV, achieving a scientific match between users' perceptual needs and the frontal shape design of automobiles, and facilitating intelligent and humanised innovation in automotive exterior design.

2.4. Automobile form generation based on SDM

The SDM is a latent diffusion model developed by the CompVis research team at the University of Munich. As a generative model based on probability and statistics in the field of deep learning, it learns the distribution characteristics of data by simulating the diffusion process of data in the noise space and its reverse denoising mechanism and then generates high-quality data samples. This model has demonstrated excellent learning and generation capabilities in multiple fields, such as image processing, audio synthesis and has become an essential part of the generative AI technology system (Liao et al. 2024).

In recent years, scholars have carried out in-depth exploration and application of SDM in different design fields. Kang and Zhao (2025) proposed a method for optimising the form design of ceramics by combining SDM with GRU-Attention. Wang, Pan et al. (2024) combined the Superiority Chart-Entropy Weight method with SDM to develop a method for rendering the design of multi-emotional products. Shi et al. (2024) proposed a generative AI architectural facade conceptual design method based on design intent.

Although SDM has achieved preliminary application results and received positive feedback in the fields of architecture and product design, research on converting a single view of a vehicle into a 3D full view is still limited. Given the significant potential of this technology in terms of shape generation and improvement of design efficiency, this study proposes an innovative approach that combines SDM with KE and delves into its practical application in automotive shape design.

3. The proposed research framework

This paper proposes a method for the form design of the front-face of NEV that combines KE and SSA-XGBoost. In the proposed research process, ChatGPT is used to collect customers' emotional data and then establish a Kansei database. Subsequently, the Entropy-AHP method is employed to obtain an accurate ranking of users' Kansei needs. After that, the SSA-XGBoost mapping model is established to transform users' needs into design features, providing new design ideas for the field of NEV design. The specific research framework is shown as follows (Figure 1).

Step 1: Systematically sort out and analyse the existing knowledge, experience and data in the automotive field using KE. With the help of ChatGPT and SDM, construct a rich emotional database and a brand-new database of NEV that have not been launched on the market, providing sufficient data support for subsequent analysis.

Step 2: Combine the Entropy-AHP method to obtain the priority ranking of Kansei words, providing a reliable decision-making basis for which customers' emotional needs should be prioritised in the automotive design process. At the same time, summarise the large number of generated front-face shape images of automobiles and establish a morphological analysis table.

Step 3: Introduce the SSA-XGBoost model to predict customers' emotional needs and automotive design features accurately. Combine the priority ranking of Kansei words and the morphological analysis table to design automotive solutions that meet customers' emotional needs, ensuring that the front-face shape design of the automobile best matches customers' emotional expectations.

Step 4: SDM generates the three-view diagrams of the car based on the design plan of the front view and builds the three-dimensional model of the car using Rhino. Subsequently,

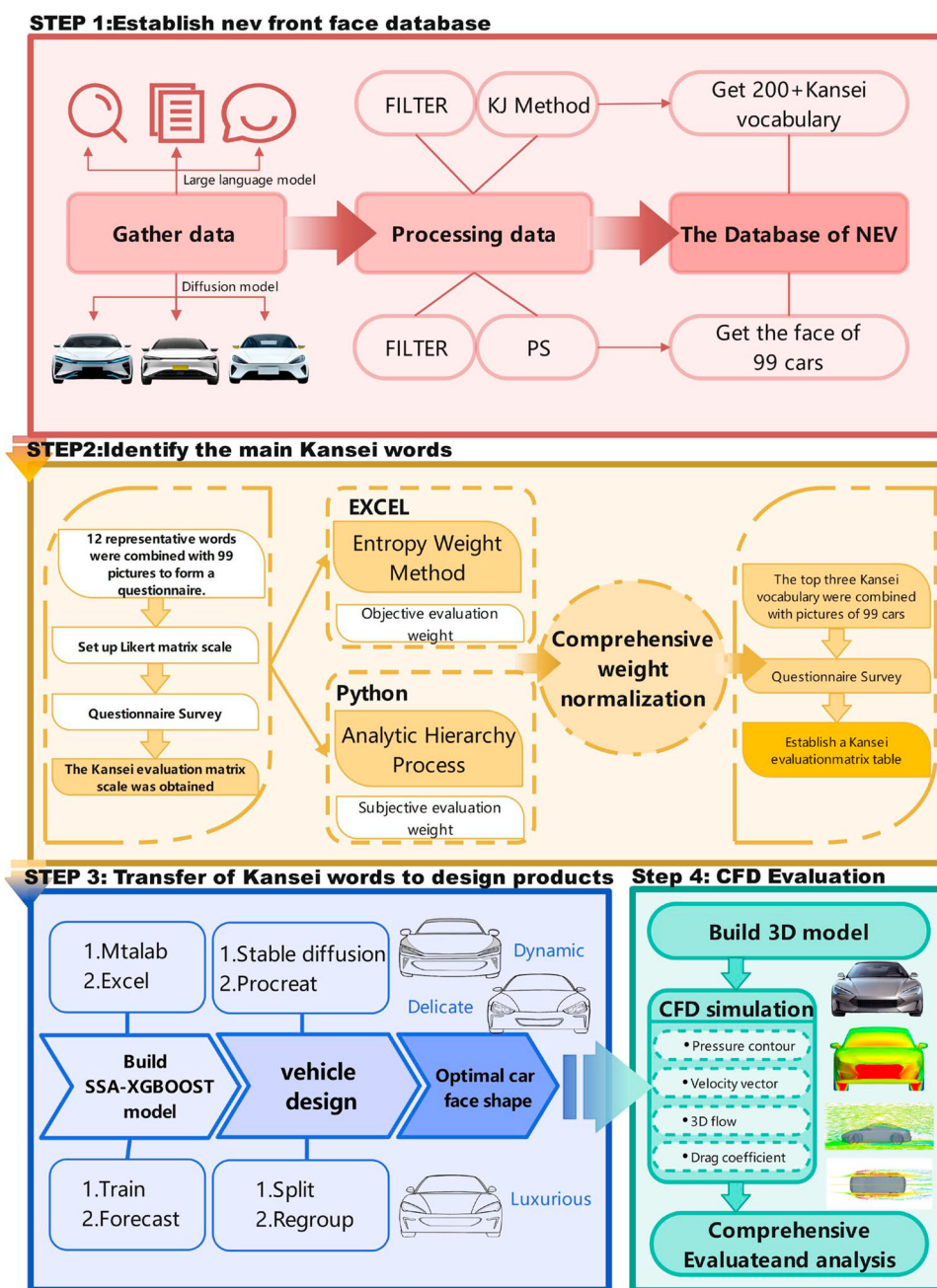


Figure 1. Proposed research framework.

tests and optimisations are conducted on the front face design of the car at both the intuitive and rational levels. At the intuitive level, expert evaluation is chosen to test and match the design results obtained with the three Kansei words in human intuitive intentions. At the rational level, the CFD software is used to conduct aerodynamic experiments on the three-dimensional model, simulating and analysing the airflow distribution, drag coefficient, etc.

3.1. ChatGPT collects NEV's Kansei words

With the development of AI, ChatGPT, a tool based on natural language processing and machine learning, performs exceptionally well in natural language processing. Its large-scale pre-training paradigm based on the Transformer architecture, through self-attention mechanisms and multi-task learning strategies, demonstrates cross-domain language understanding and generation capabilities. ChatGPT excels in aspects such as sentiment analysis, text generation and exhibits characteristics similar to human thought chains in complex reasoning tasks. Compared to traditional surveys, it has significant advantages in terms of time, manpower and resources. In terms of the time dimension, traditional surveys require multiple steps such as questionnaire design, sample sampling, and a single research cycle often lasts for several weeks. However, ChatGPT can complete multiple rounds of data generation and preliminary analysis within minutes. In terms of cost, traditional surveys involve multiple expenditures such as manpower, materials, incentives and the investment often amounts to tens of thousands of yuan. ChatGPT only requires basic service fees and can significantly reduce economic costs. In terms of data output efficiency, traditional surveys are limited by sample size and frequency and data accumulation is slow. ChatGPT, on the other hand, can generate index-level text data based on themes and rapidly extract key emotional elements through multiple iterations, with data generation efficiency increasing by nearly a hundred times. Thus, ChatGPT breaks through the limitations of traditional research in terms of samples and regions, achieving comprehensive coverage and in-depth exploration of users' emotional needs, fully demonstrating the core value of AI in enhancing the efficiency of emotional engineering research.

In automotive design, we can quickly obtain a large amount of effective databases through large-language models. Firstly, we can have a conversation with the large language model to get adjectives related to the front-face shape of automobiles. The dialogues are as follows: 'You are now a senior and excellent product designer. Please list the Kansei words for the front-face shape design of automobiles'. 'You are now a consumer who wants to buy a NEV. Please list the Kansei words to describe the front-face shape of your ideal car model'. The Kansei words generated from different perspectives can help designers understand market demands more comprehensively and design higher-quality automotive products that meet market expectations.

Therefore, this study takes advantage of the large language model ChatGPT to generate a large-scale Kansei words database required for the research. Compared with traditional KE data acquisition methods, it is faster and can be iteratively optimised repeatedly, thus improving the efficiency of database acquisition.

3.2. SDM generates new NEV image data

Input the large amount of effective Kansei dataset generated and organised by ChatGPT in the previous stage into the SDM. Set the corresponding parameters according to the target to obtain the required dataset of hundreds of images.

First, the user inputs a text describing the target image, for example, 'A car, realistic, the overall presentation of the luxury texture of the car face, 3D, clear rendering'. This text is input into the model as important conditional information.

Then, enter the forward diffusion stage. In the latent space, starting from an initial random noise image, Gaussian noise is gradually added to it according to the preset diffusion

steps, causing the image to gradually evolve into a completely random noise distribution state. This is the ‘destruction’ of the data, laying the foundation for subsequent reverse generation. During the forward diffusion process, noisy images at different time steps are obtained by gradually adding Gaussian noise to the initial image. The formula is as follows:

$$q(x_{1:T}|x_0) = \prod_{t=1}^T q(x_t|x_{t-1}) \quad (1)$$

$$q(x_t|x_{t-1}) = N\left(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I\right) \quad (2)$$

x_0 is the original image and x_t is the image after T-step noise processing.

Then comes the crucial reverse generation stage. The model uses a deep neural network (such as the U-Net architecture) and the knowledge learned from a large amount of data through self-supervised and unsupervised learning. Combining the input text information, it starts to gradually restore the image from the noisy state in the latent space.

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}z_t, \bar{\alpha}_t = \alpha_t\alpha_{t-1}\alpha_{t-2} \quad (3)$$

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1}|x_t) \quad (4)$$

$$p_\theta(x_{t-1}|x_t) = N\left(x_{t-1}; \mu_\theta(x_t, t), \sum_{\theta} (x_t, t)\right) \quad (5)$$

To better understand the steps of generating images by SDM, the process is expressed using a visual chart (Figure 2). During the SDM training phase, the reconstruction loss part needs to be calculated to measure the pixel-level mean square error between the generated car image and the real car image, minimising the difference between the two to ensure that the generated image achieves high fidelity in terms of contour, details and lighting. Secondly, by constraining the relative Entropy between the generated distribution and the preset prior distribution, the diversity of the generated car image can be dynamically adjusted, effectively avoiding the problems of mode collapse and overfitting. Under the collaborative optimisation of this composite loss function, SDM achieves a dynamic balance between the authenticity and innovation of car images.

By conditionally modifying the internal diffusion model architecture, SDM acquires the core capability of generating car images from text prompts. In the text processing process, the CLIP text encoder is frozen as a feature extractor, which can efficiently map ‘exquisite’, ‘NEV’, etc., natural language descriptions, into semantic feature vectors. These semantic vectors serve as conditional inputs and are deeply coupled with the iterative generation process of the diffusion model, guiding the model to generate high-quality car images with corresponding styles and forms, significantly improving the efficiency and controllability of car image creation. When a large-scale dataset rich in multi-dimensional information such as car shape features, styles, elements, etc. is input into SDM, the model, based on machine learning algorithms and text semantic constraints, ultimately outputs diverse styles and rich details of car images.

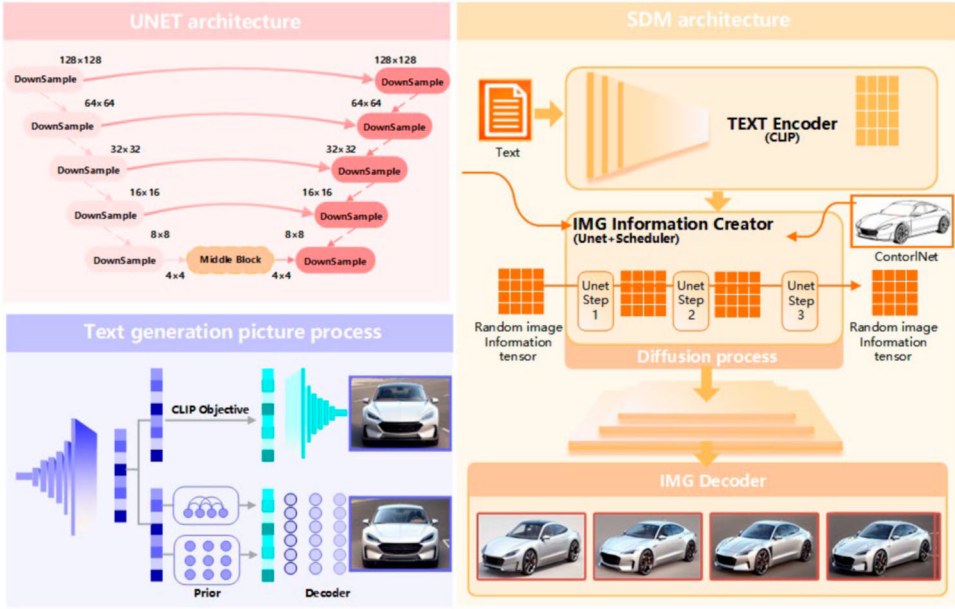


Figure 2. The SDM generation process.

3.3. Entropy-AHP calculates Kansei weights

3.3.1. Entropy

When performing Entropy calculations, since the measurement units of various indicators in the dataset are not uniform, the first step before calculating their comprehensive indicators is to standardise the data. This process eliminates the differences in dimensions and transforms the original data into a dimensionless standardised matrix.

- (1) Positive indicators

$$y_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} \quad (6)$$

Make sure that the indicator values are mapped to the range [0, 1], the larger the value, the better.

- (2) Negative indicators:

$$y_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \quad (7)$$

Reverse scaling ensures that the larger the value, the better.

- (3) Specific gravity calculation

$$p_{ij} = \frac{y_{ij}}{\sum_{i=1}^m y_{ij}} \quad (8)$$

(4) Information entropy calculation

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^m p_{ij} \ln p_{ij} \quad (9)$$

(5) Calculation of difference coefficient

$$d_j = 1 - e_j \quad (10)$$

(6) Weight calculation

$$w_j = \frac{d_j}{\sum_{j=1}^n d_j} \quad (11)$$

3.3.2. AHP

(1) Establish a hierarchical structure model.

This paper uses the AHP model, which is divided into three levels: target layer, criterion layer and sub-criterion layer.

(2) Establish a judgment contrast matrix.

The judgment matrix is the basic component of AHP. In it, elements of the same level need to be compared pairwise and given a certain value. Generally, nine scales from 1 to 9 are used as the judgment scale of elements in the judgment matrix. By judging each two elements of the matrix one by one according to the scale, the comparison judgment matrix A of order n can be obtained. N-order judgment matrix:

$$A_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{1..} & a_{1n} \\ a_{21} & a_{22} & a_{2..} & a_{2n} \\ a_{..} & a_{..} & a_{..} & a_{..} \\ a_{n1} & a_{n2} & a_{n..} & a_{nn} \end{bmatrix} \quad (12)$$

(3) Merge the expert matrices. Using the geometric mean method can ensure the consistency check. Multiply the scoring matrices formed by m experts ($m = 1, 2, \dots, k$) element-by-element, and then take the m -th root to obtain the unique integrated matrix $\bar{A}$. The formula is as follows:

$$\bar{A} = \left(\prod_{k=1}^m a_{ij}^k \right)^{\frac{1}{m}} \quad (13)$$

(4) Calculate the relative weights of the unique integrated matrix.

The most commonly used methods for calculating the maximum eigenvector of the pairwise comparison matrix are the sum-product method and the root-square method. The sum-product method is adopted in this paper. The specific calculation steps of the sum-product method are as follows:

1. Obtain the unique integrated matrix $\mathbf{A}$.
2. Normalise each column vector of matrix $\mathbf{A} = (a_{ij})_{n \times n}$ to obtain matrix $\mathbf{B} = (b_{ij})_{n \times n}$:

$$b_{ij} = \frac{a_{ij}}{\sum_{i=1}^n a_{ij}} \quad (14)$$

3. Sum the row vectors of the normalised matrix $\mathbf{B}$:

$$M_i = \sum_{j=1}^n b_{ij} \quad (15)$$

$$4. \text{ Normalise the vector } M = \begin{bmatrix} M_1 \\ M_2 \\ \dots \\ M_n \end{bmatrix}$$

$$W_i = \frac{M_i}{\sum_{i=1}^n M_i} \quad (16)$$

$$\text{Obtain the eigenvector } W_i = \begin{bmatrix} W_1 \\ W_2 \\ \dots \\ W_n \end{bmatrix}$$

5. Calculate the maximum eigenvalue λ_{max} :

$$\lambda_{max} = \frac{1}{n} \sum_{i=1}^n \frac{[AW]_i}{W_i} \quad (17)$$

6. Determine whether the matrix passes the consistency check.

- (a) (1) Calculate the consistency index:

$$CI = \frac{\lambda_{max} - n}{(n - 1)} \quad (18)$$

- (b) (2) Calculate the consistency ratio:

$$CR = \frac{CI}{RI} = \frac{\lambda_{max} - n}{(n - 1)RI} \quad (19)$$

Calculate the consistency ratio: $CR = \frac{CI}{RI}$. When $CR \leq 0.1$, it is considered that the judgment matrix has satisfactory consistency. Otherwise, the judgment matrix needs to be adjusted.

3.3.3. Entropy-AHP normalise the comprehensive weights

1. Determine the comprehensive weights.

Let the weight vector obtained by the Entropy method be $W^e = (w_1^e, w_2^e, \dots, w_m^e)$, Let the weight vector obtained by the AHP method be $W^a = (w_1^a, w_2^a, \dots, w_m^a)$, the comprehensive weight $W = (w_1, w_2, \dots, w_m)$ is determined by means of linear weighting. The

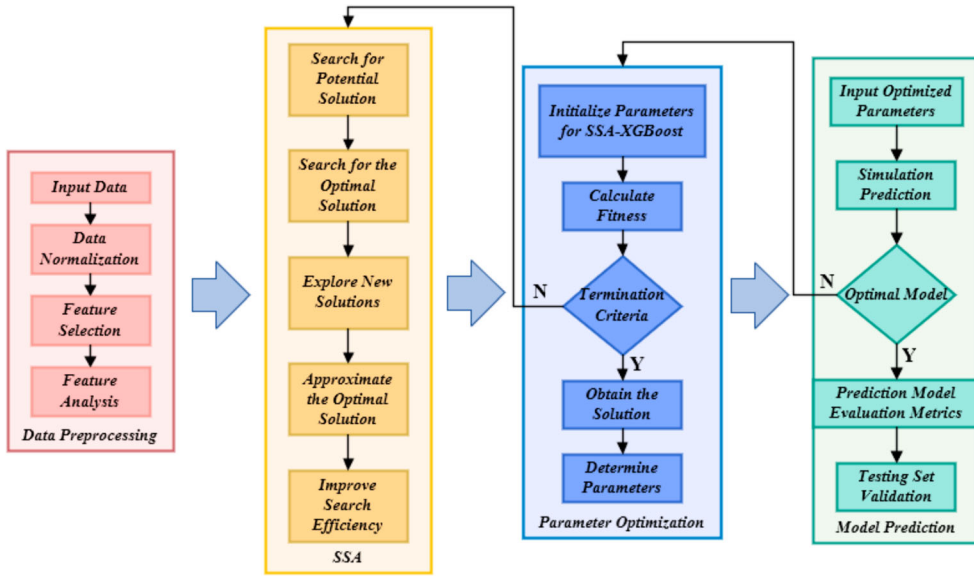


Figure 3. Structure of the SSA-XGBoost model.

formula is $w_j = \alpha w_j^e + (1 - \alpha)w_j^a$, where is α the adjustment coefficient, $0 \leq \alpha \leq 1$. Due to the need for comprehensive consideration of the data, α was set to 0.5, meaning that the entropy weight method and the analytic hierarchy process method would each have a 50% weight for calculating the comprehensive weight, thereby achieving a comprehensive assessment of the effects of the two methods.

2. Weight normalisation.

Normalise the comprehensive weight W . The formula is:

$$\bar{w}_j = \frac{w_j}{\sum_{i=1}^m w_j} \quad (20)$$

Obtain the normalised comprehensive weights:

$$\bar{W} = (\bar{w}_1, \bar{w}_2, \dots, \bar{w}_m) \quad (21)$$

3.4. Establish the Kansei mapping model with SSA-XGBoost

The SSA-XGBoost model (Figure 3), with its unique architectural and algorithmic advantages, is capable of keenly and accurately capturing complex nonlinear dependencies and the interactions among multi-dimensional perceptual features.

XGBoost is an efficient and powerful machine learning algorithm based on the gradient boosting framework. The first step is to find the objective function, which includes a loss function and a regularisation term:

Objective function: It includes a loss function that measures the difference between the predicted values and the true values, and a regularisation term that controls the complexity

of the model.

$$Obj(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (16)$$

Basis Function (Decision Tree): The decision tree partitions the sample space into T leaf nodes, where w_j represents the score of the leaf node, and $q(x)$ denotes the index of the leaf node to which the sample belongs.

$$f_k(x) = w_{q(x)} \quad (17)$$

Model training: At the T -th iteration, the leaf node fraction w_j of the new basis function $f_t(x)$ is obtained by solving. g_i and h_i represent the first and second-order derivatives of the loss function, respectively and λ is the regularisation parameter.

$$\operatorname{argmin}_w \sum_{i=1}^n \left[g_i w_{q(x_i)} + \frac{1}{2} h_i w_{q(x_i)}^2 \right] + \lambda \sum_{j=1}^T w_j^2 \quad (18)$$

SSA is used to improve the search speed for the optimal solution in XGBoost. The first step is to randomly initialise the positions of the sparrow population, where the population size is N , and each sparrow's position is a dd -dimensional vector representing a set of solutions to the optimisation problem. The position matrix of the sparrow population X can be expressed as:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,d} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,d} \end{bmatrix} \quad (19)$$

The second step is the update of the discoverer's position:

$$X_{ij}^{t+1} = \begin{cases} X_{ij}^t \cdot \exp\left(-\frac{i}{\alpha T}\right), & R_2 < ST \\ X_{ij}^t + Q \cdot L, & R_2 \geq ST \end{cases} \quad (20)$$

The third step is the update of the participant's position. The participant follows the discoverer to obtain the food. The formula for the position update is as follows:

$$X_{ij}^{t+1} = \begin{cases} Q \cdot \exp\left(\frac{X_{worst}^t - X_{ij}^t}{i^2}\right), & i > \frac{N}{2} \\ X_{pj}^{t+1} + |X_{ij}^t - X_{pj}^{t+1}| \cdot A^+ \cdot L, & i \leq \frac{N}{2} \end{cases} \quad (21)$$

The fourth step is the update of the alert's position:

$$X_{ij}^{t+1} = \begin{cases} X_{best}^t + \beta \cdot |X_{ij}^t - X_{best}^t|, & f_i > f_g \\ X_{ij}^t + K \cdot \left(\frac{|X_{ij}^t - X_{worst}^t|}{(f_i - f_w) + \varepsilon} \right), & f_i = f_g \end{cases} \quad (22)$$

Finally, the fitness value of each sparrow is calculated after updating their positions, and the global best and worst solutions are updated. The algorithm checks whether the maximum number of iterations has been reached. If so, the algorithm terminates and outputs the optimal solution.

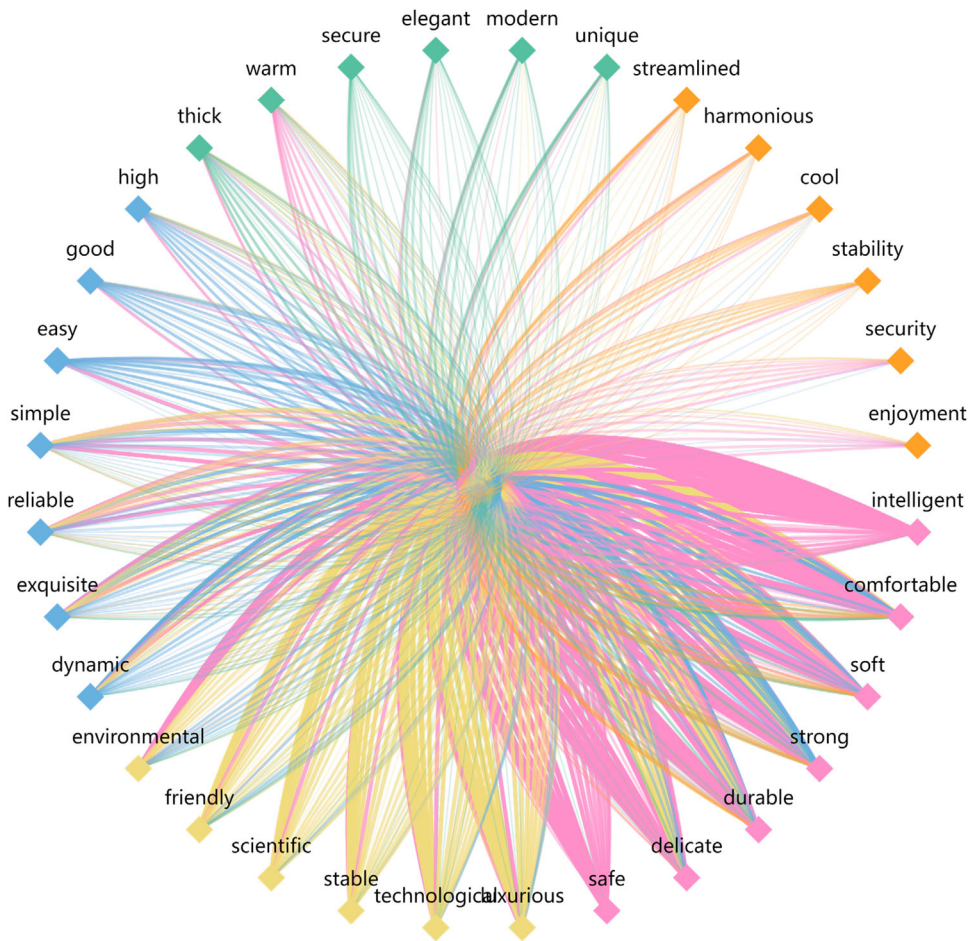


Figure 4. Kansei vocabulary dataset semantic network.

4. Case study

4.1. Kansei words collection

Due to the slow speed and low efficiency of traditional information collection methods, this study utilises ChatGPT to collect and organise Kansei words.

Step 1: Input instructions to ChatGPT-4.0, asking it to act as a senior automotive designer and list 200 imagery adjectives that can describe the front shape of an automobile. Screen the generated words for professionalism and accuracy and finally determine 200 words as the Kansei word data for the front-face shape design of NEV (Figure 4). Step 2: Use the KJ method to conduct a cluster analysis on the 200 determined Kansei words describing the front-face shape of automobiles. Then, sort out 12 representative Kansei words, which are Comfortable, Luxurious, Intelligent, Reassuring, Energy-efficient, Environmental friendly, Exquisite, Durable, Dynamic, Steady, Fluent, and Safe.



Figure 5. Car front-face dataset.

4.2. Collection of NEV frontal facial form images

Input the organised Kansei words dataset into the SDM and generate hundreds of novel front-face shapes of NEV through the trained model (Lora). First, train the processed data into Lora to control the stability and accuracy of the front-face shape of NEV during the production process. After the training is completed, match the trained Lora model with the accurate geometric large-scale model and generate 300 images in the SDM software. After eliminating the pictures that do not meet the standards, invite 10 automobile-related experts to select 99 high-quality front-face shape images of NEV as samples from the generated images through group discussion; the screening process includes removing invalid and repetitive data. Then, use Photoshop to remove the background and add a white background. Use the Image Tuner software to uniformly adjust the images to 128*128 pixels and convert them to the JPEG format. Finally, a high-quality dataset containing 99 front-face shape design images of automobiles is successfully constructed (Figure 5).

4.3. Deconstruction of NEV's front-face form

A focus group consisting of five industrial design experts was selected. Using the morphological analysis method, the front-face shape of NEV was decomposed into seven categories and five representative shapes were summarised for each category. The seven categories are headlights, fog lights, engine hood, air intake grille, rearview mirrors, windshield and

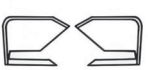
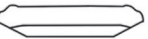
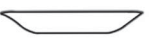
Design features	Type 1	Type 2	Type 3	Type 4	Type 5
Headlights					
Fog lights					
Hood					
Intake grille					
Rearview mirror					
Windshield					
Contour map					

Figure 6. Morphological analysis table.

body contour. The five representative shapes were obtained through summarisation and induction, with each representing a different design style and detail (Figure 6).

4.4. Entropy-AHP ranking key Kansei weights

To establish the mapping relationship between customers’ emotions and the car’s front face design, it is necessary to first sort the key Kansei words. First, 99 images were combined with 12 Kansei words to create a seven-level Likert scale questionnaire. Besides the consumers’ emotional intentions regarding the design of the car’s frontal shape, we also collected information such as the age, gender, occupation and income of the participants (Figure 7). The final questionnaire reliability was 0.79, indicating that the questionnaire is highly reliable and the data is valid. This study used a combination of subjective and objective methods, namely Entropy-AHP, to sort the Kansei words based on the questionnaire data.

4.4.1. Entropy calculated weight

The 12 Kansei words are objectively weighted and ranked according to the Entropy formula, which is divided into four steps:

In the first step, dimensionless and non-negative processing results can be obtained according to formulas (6–7) (Table 1).

Table 1. Dimensionless and nonnegative treatment results Y_{ij} .

Y_{ij}	Vehicle 1	Vehicle 2	Vehicle 3	Vehicle 4	Vehicle 5	Vehicle 6	Vehicle 7	Vehicle 8	...	Vehicle 99
Environmentally friendly	0.80	1.00	0.60	0.60	0.60	0.60	0.60	0.80	...	1.00
Intelligent	1.00	0.60	0.40	1.00	0.60	0.80	0.80	0.80	...	0.60
Energy efficient	0.60	0.80	0.60	0.40	0.80	0.80	0.80	0.80	...	0.40
Safe	0.80	0.80	0.40	0.80	0.80	0.80	0.60	0.80	...	0.60
Luxurious	0.33	0.67	0.83	0.67	0.67	0.67	0.67	0.67	...	0.67
Soft	0.67	0.83	0.83	0.83	0.83	0.83	1.00	0.83	...	0.67
Dynamic	0.60	0.60	0.80	0.80	0.80	0.80	0.80	0.60	...	0.60
Exquisite	0.60	0.40	0.40	0.80	0.60	0.80	1.00	0.60	...	1.00
Comfortable	0.60	0.80	0.60	0.80	0.80	0.80	0.60	0.60	...	0.60
Reassuring	0.83	0.50	0.67	0.83	0.67	0.83	1.00	0.83	...	0.67
Steady	0.60	0.80	0.60	0.60	0.80	0.60	0.40	1.00	...	0.80
Fluent	0.60	0.80	0.40	0.80	0.60	0.80	0.80	1.00	...	0.80

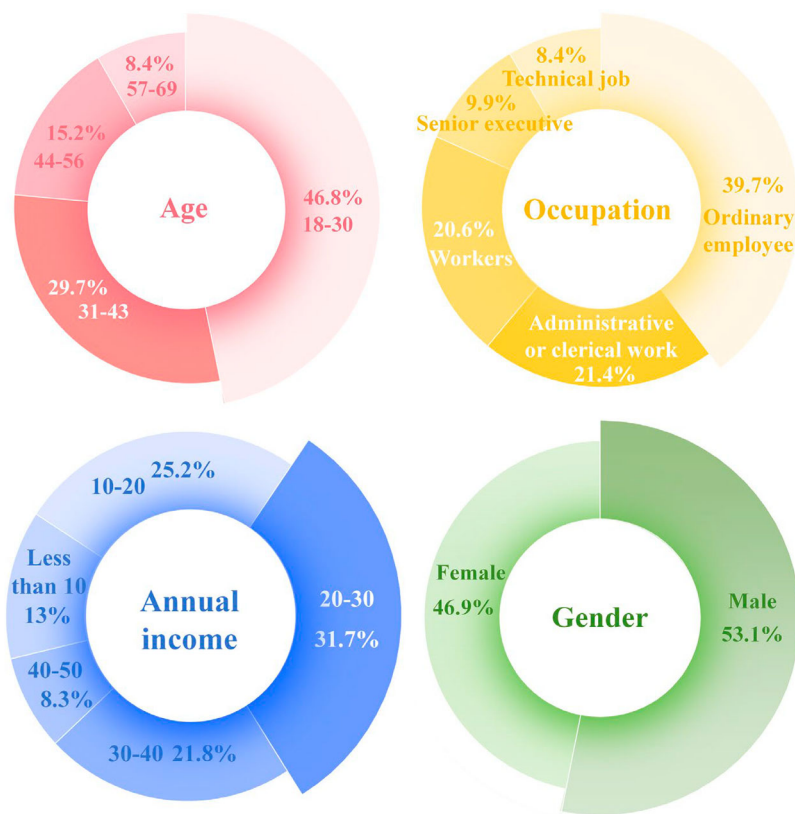


Figure 7. The age, occupation, annual income and gender of the research subjects.

In the second step, the specific gravity matrix can be obtained according to formula (8) (Table 2).

In the third step, information entropy and difference coefficient can be obtained according to the formulas (9–10) (Table 3).

The fourth step is to calculate the objective weights of each index according to formula (11) (Table 4).

4.4.2. AHP calculation weight

Python software is used to objectively assign and sort 12 Kansei words through AHP in this stage. The specific steps are as follows: First, the target layer, criterion layer and sub-criterion layer are determined. The target layer is the design scheme of the face shape. The criterion level is the functional level, form level and driving sense level. The sub-criterion layer is 12 Kansei words in the data set: Comfortable, Luxurious, Intelligent, Reassuring, Energy-efficient, Environmental-friendly, Exquisite, Durable, Dynamic, Steady, Fluent.

To ensure the accuracy of index weight calculation, 10 experienced automotive product designers were invited to score the indicators at each criterion level. According to the 1–9 scale method, pairwise comparisons were made among the three indicators at the criterion level, and then pairwise comparisons were carried out among the 12 sub-criteria. Thus, the

Table 2. Proportion of each index in each scheme P_{ij} .

P_{ij}	Vehicle 1	Vehicle 2	Vehicle 3	Vehicle 4	Vehicle 5	Vehicle 6	Vehicle 7	Vehicle 8	...	Vehicle 99
Environmentally friendly	0.0018	0.0023	0.0014	0.0014	0.0014	0.0014	0.0014	0.0018	...	0.0023
Intelligent	0.0025	0.0015	0.0010	0.0025	0.0015	0.0020	0.0020	0.0020	...	0.0015
Energy efficient	0.0015	0.0020	0.0015	0.0010	0.0020	0.0020	0.0020	0.0020	...	0.0010
Safe	0.0020	0.0020	0.0010	0.0020	0.0020	0.0020	0.0015	0.0020	...	0.0015
Luxurious	0.0008	0.0016	0.0020	0.0016	0.0016	0.0016	0.0016	0.0016	...	0.0016
Soft	0.0016	0.0020	0.0020	0.0020	0.0020	0.0020	0.0024	0.0020	...	0.0016
Dynamic	0.0015	0.0015	0.0021	0.0021	0.0021	0.0021	0.0021	0.0015	...	0.0015
Exquisite	0.0015	0.0010	0.0010	0.0020	0.0015	0.0020	0.0025	0.0015	...	0.0025
Comfortable	0.0015	0.0020	0.0015	0.0020	0.0020	0.0020	0.0015	0.0015	...	0.0015
Reassuring	0.0020	0.0012	0.0016	0.0020	0.0016	0.0020	0.0024	0.0020	...	0.0016
Steady	0.0015	0.0020	0.0015	0.0015	0.0020	0.0015	0.0010	0.0025	...	0.0020
Fluent	0.0015	0.0020	0.0010	0.0020	0.0015	0.0020	0.0020	0.0025	...	0.0020

Table 3. Index entropy result E_j and G_j .

Kansei words	E_j	G_j
Environmentally friendly	0.991	0.008
Intelligent	0.990	0.009
Energy efficient	0.988	0.011
Safe	0.990	0.009
Luxurious	0.993	0.006
Soft	0.993	0.006
Dynamic	0.989	0.010
Exquisite	0.990	0.009
Comfortable	0.988	0.011
Reassuring	0.993	0.006
Steady	0.990	0.009
Fluent	0.991	0.008

Table 4. Indicator weight result W_j .

Kansei words	W_j
Environmentally friendly	0.075
Intelligent	0.085
Energy efficient	0.111
Safe	0.090
Luxurious	0.064
Soft	0.059
Dynamic	0.095
Exquisite	0.088
Comfortable	0.105
Reassuring	0.057
Steady	0.087
Fluent	0.078

Table 5. Weights determined by AHP.

Primary index	First level weight	Secondary indicators	Secondary weight	Comprehensive weight
Functional hierarchy	0.2981	Environmentally friendly	0.102	0.030
		Intelligent	0.343	0.102
		Energy efficient	0.237	0.070
		Safe	0.316	0.094
Morphological hierarchy	0.4829	Luxurious	0.294	0.142
		Soft	0.108	0.052
		Dynamic	0.243	0.117
		Exquisite	0.354	0.171
Driving sense level	0.219	Comfortable	0.378	0.082
		Reassuring	0.186	0.040
		Steady	0.197	0.043
		Fluent	0.237	0.052

judgment and evaluation matrix were obtained. Through the formulas (12–19), the weight of each index, the consistency index and the test coefficient were calculated (Table 5).

4.4.3. Entropy-AHP comprehensive weight normalisation

Obtain the comprehensive weights of Entropy AHP using formulas (20–21). The top three adjectives in comprehensive ranking are Exquisite, Dynamic and Luxurious (Table 6).

Table 6. Entropy-AHP analysis process.

	The comprehensive weight calculated by Entropy-AHP	Comprehensive weight normalisation	The comprehensive weight after normalisation
Environmentally friendly	0.027	0.084	0.025
Intelligent	0.102	0.317	0.095
Energy efficient	0.093	0.288	0.086
Safe	0.100	0.311	0.093
Luxurious	0.108	0.238	0.115
Soft	0.037	0.081	0.039
Dynamic	0.131	0.289	0.139
Exquisite	0.179	0.392	0.189
Comfortable	0.102	0.459	0.101
Reassuring	0.028	0.125	0.027
Steady	0.044	0.199	0.044
Fluent	0.048	0.216	0.047

Table 7. Kansei evaluation matrix.

NO.	Headlight	Fog lights	Hood	Intake grille	Rearview mirror	Windshield	Contour map	Luxurious	Exquisite	Dynamic
1	5	1	4	3	2	2	5	5.28	5.19	5.11
2	5	4	5	2	2	2	5	5.47	5.10	5.33
3	5	1	4	3	1	5	5	5.47	5.08	5.24
4	5	2	3	3	1	3	3	5.24	5.28	5.08
5	3	5	2	1	3	2	3	5.09	5.47	5.06
6	3	1	3	2	4	3	3	5.01	5.51	4.94
7	5	2	2	4	1	2	5	5.24	5.32	5.12
8	4	1	2	3	3	5	3	5.15	5.32	5.10
...	...	...	...	...	...	...	...	...	...	...
98	4	2	5	3	2	2	4	5.29	5.27	5.18
99	1	5	2	4	5	5	4	5.52	5.07	5.29

4.4.4. Establish Kansei evaluation matrix

To establish the Kansei evaluation matrix, a questionnaire was created using 99 representative front-face design schemes of automobiles and three key Kansei adjectives obtained from the Entropy-AHP method. Then, after using the 7th-order Likert scale, the Kansei evaluation matrix was formed (Table 7).

4.5. SSA-XGBoost establishes a mapping association

Import the data from Table 7 into the SSA-XGBoost for Kansei mapping analysis. The data composed of 99 NEV are divided into a training set and a test set at a ratio of 70% and 30%. Among them, the SSA-XGBoost prediction model optimises the extreme gradient boosting tree with SSA and introduces a feature dynamic screening mechanism to improve the model performance significantly. During training, the Adadelta optimiser is used to update the parameters, the maximum number of iterations is set to 1200 and the initial learning rate is 0.01. The cosine annealing strategy is adopted for learning rate adjustment, which dynamically adjusts according to the training process to improve the convergence stability of the model. The data are randomly shuffled in each round of training and the batch size is set to 128. The validation set is independent of the training set to monitor the model

performance in real time and prevent overfitting. The training process is set to silent mode with concise output. The specific operation steps are as follows:

First, use the SSA to optimise the hyperparameters of the XGBoost model. In the initialisation stage of the algorithm, a certain number of sparrow individuals are randomly generated as the initial population. Each individual represents a set of hyperparameter combinations for XGBoost, such as the learning rate, the number of trees, and the maximum depth. During the iteration process, the population is divided into discoverers and followers. Discoverers explore new hyperparameter regions in a more extensive scope based on their fitness values, while followers adjust their positions according to the positions of discoverers and their own experience. At the same time, an early-warning mechanism is introduced. When some sparrows sense danger, some of them will quickly adjust their positions to avoid getting trapped in local optima. After searching for the optimal hyperparameter combination through multiple rounds of iteration, it lays the foundation for the training of the XGBoost model.

Second, construct the XGBoost model based on the optimised hyperparameters. Before model training, a feature dynamic screening mechanism is applied. During the training process, the feature importance scores are calculated in real time according to the contribution of each feature to the target variable. For example, the built-in gain calculation method of XGBoost is used to evaluate the impact of each feature on the objective function when splitting nodes. After a certain number of training rounds, a threshold for the importance score is set and features with scores lower than the threshold are removed. Only the key features are retained for subsequent training to improve the training efficiency and generalisation ability of the model. The XGBoost model uses decision trees as base learners and continuously iteratively trains through gradient boosting to combine multiple weak learners into a strong learner. During the training process, the second-order Taylor expansion is used to approximate the loss function, which enables a more accurate calculation of gradients and accelerates the convergence of the model. Visualise the training process and structural framework of the model (Figure 3) to facilitate a deeper understanding of the model's performance and operating mechanism.

The experimental results show that the SSA-XGBoost model demonstrates extremely excellent prediction and fitting accuracy for the target variable. Its R^2 is as high as 0.96, and the RMSE is as low as 0.04. This indicates that the SSA-XGBoost model has a good fitting effect on the training data and can capture most of the patterns and features in the data. As a result, it can relatively accurately describe the relationship between the independent and dependent variables, with a high probability of outputting prediction results close to the actual situation, providing a reliable basis for subsequent analysis and decision-making. Meanwhile, the prediction error of the model is minimal. The maximum mean absolute error (MAE) is 0.34, and the fluctuation of the predicted values is very weak. The prediction results are highly stable, which can reduce the risks and losses caused by excessive prediction errors and improve the scientific nature and accuracy of decision-making. Therefore, it can be inferred that the SSA-XGBoost model has stable and accurate prediction performance, showing good adaptability and reliability (Table 8).

After the SSA-XGBoost model establishes the mapping relationship, it predicts the Kansei evaluation values of all $5 \times 7 = 78,125$ groups of three Kansei words by learning the high-score features of the input data. By arranging the scores from high to low, the optimal combinations corresponding to the three high-score adjectives 'Exquisite', 'Dynamic' and

Table 8. SSA-XGBoost model training set and test set data.

	R^2	RMSE	MAE
Train	0.96	0.04	0.32
Test	0.95	0.04	0.34

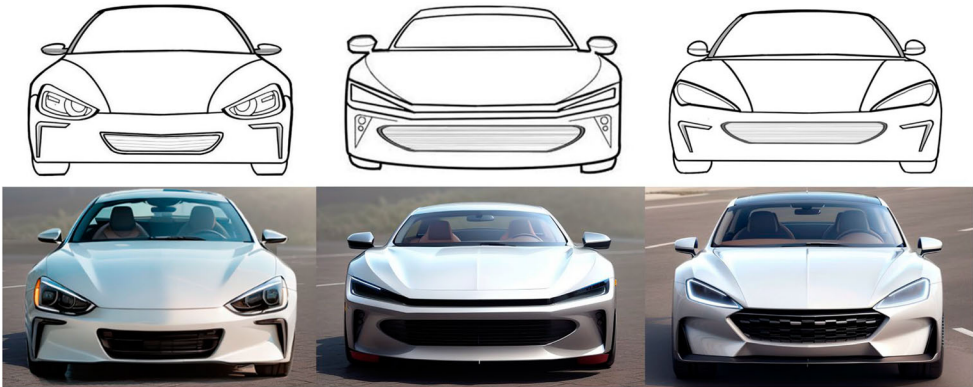


Figure 8. ‘Exquisite, Dynamic, Luxurious’ black and white drafts and renderings.

‘Luxurious’ can be determined, which are (4, 4, 5, 1, 4, 5, 3), (1, 1, 2, 2, 1, 1, 2) and (5, 4, 5, 4, 5, 5, 3) respectively. According to the form analysis table combination and design, obtain the high score car face shape design in line with the three emotional words and draw the black and white line draft by hand.

4.6. SDM generated rendering scheme

To convert the black-and-white line drawing into a high-quality car rendering image (Figure 8), we carried out the creative process using the text-to-image and image-to-image functions in the Liblib platform, which is based on the SDM model as its core technology (Figure 9). The specific parameter adjustment process is as follows:

Firstly, in terms of sampler selection, Euler a was chosen as the preferred option due to its efficient and stable characteristics. By optimising the iterative process, it significantly improves the generation efficiency while ensuring the quality of the output images. After testing, setting the sampling frequency within the range of 30 to 40 steps allows for a good balance between speed and accuracy. Subsequently, based on the actual results, adjustments can be made flexibly to obtain the desired design images of the car’s frontal shape.

Secondly, for the key parameter Classifier Free Guidance (CFG) scale, its essence is to regulate the degree of constraint that the prompt words impose on the generated result. When the value is low, the model has a higher degree of creative freedom, but it is prone to generating content that deviates from the prompt; when the value is too high, the image will be overly constrained by the prompt, resulting in a stiff image and missing details. Through multiple rounds of testing, it was found that controlling it within the range of 3.5 to 5 can ensure that the generated image closely matches the core prompts such as

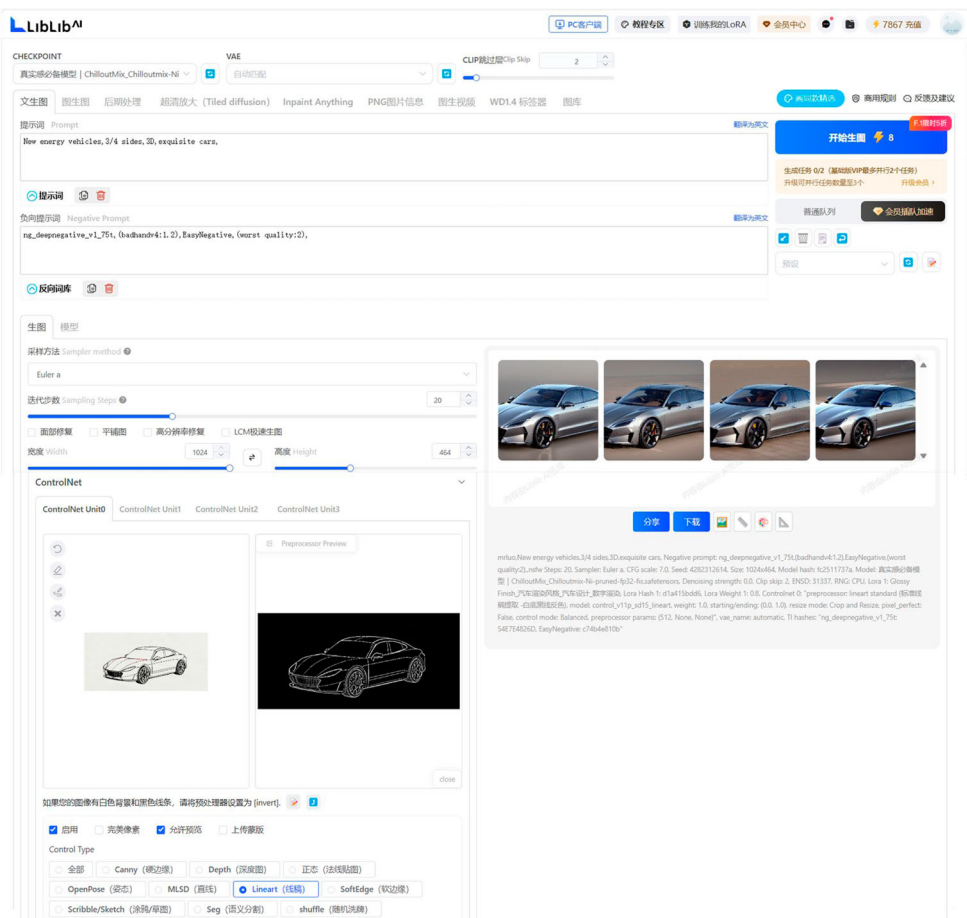


Figure 9. Liblib SDM software generation process.

‘New energy vehicles, 3/4 sides, 3D, exquisite cars’, while also retaining the model’s creative space, achieving the best balance between creative expression and semantic matching.

Furthermore, to enhance the consistency of the generated images with the original line drawings in terms of style, we introduced ControlNet for auxiliary control. By precisely setting the control weights and guiding parameters, we ensured that the main structure of the car strictly followed the contour of the line drawing. At the same time, we set the CLIP skip layer parameter to 2. This adjustment can moderately weaken the detail constraints in the text prompts, retaining only the core semantic information. Thus, the model, while accurately capturing key concepts such as ‘black-and-white line drawing of the car front view’, can more flexibly interpret ‘exquisite’ and ‘dynamic’ etc. descriptive terms, effectively avoiding the problem of rigid generation caused by excessive semantic subdivision. With ControlNet’s stable control of the main structure, the model can fully exert its design creativity and ultimately output high-quality car renderings (Figure 10). By importing the obtained images and the input keyword text into Python, the model calculates the Clip score. The scores for the three words ‘exquisite’, ‘dynamic’ and ‘luxurious’ are 0.31, 0.29,



Figure 10. Iteration diagrams generated by SDM.

and 0.31 respectively. This clearly indicates that the model performs better than most of the existing models in understanding the text and generating the output.

4.7. CFD

After completing the mapping between customers' Kansei needs and product design features and screening the finally generated schemes, an expert group was established to conduct a comprehensive evaluation of the automobile design. Leveraging the experts' rich professional knowledge in automobile exterior design, a second-round assessment was carried out on the three pre-screened image schemes. Since in the previous calculation results, the word 'Luxurious' had the highest comprehensive Kansei score, indicating the most substantial emotional needs of users, the 'Luxurious' form combination was ultimately selected for CFD. The Rhino modelling software established a three-dimensional digital model based on the three-view drawings generated by the SDM (Figure 11), and the CFD method was introduced to evaluate the feasibility of the selected scheme.

In the field of automotive aerodynamics, CFD, as a powerful numerical analysis tool, can evaluate various parameters such as the velocity vector of the vehicle, the pressure cloud on the vehicle body surface, the residual wind resistance coefficient, the velocity vector at the mid-section and 3D curves. Through the data of these indicators, the performance of the design results can be evaluated and the feasibility of the design scheme can be verified. The automotive geometric model constructed by Rhino is imported into ANSA to divide the mesh, the areas with significant changes in the body and the flow field are densely processed (Figure 11).

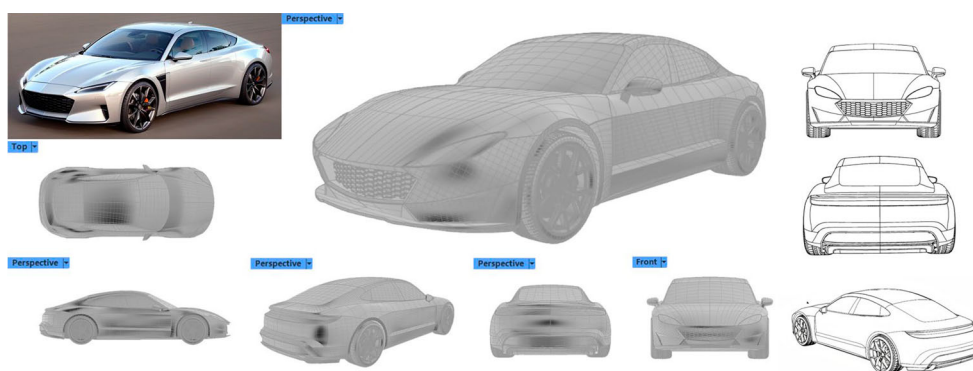


Figure 11. 3D models and creative sketches.

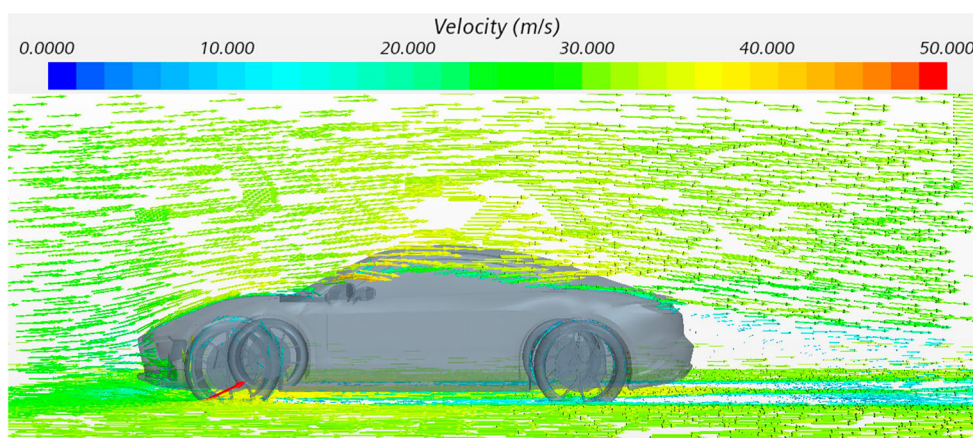


Figure 12. Velocity vector diagram.

The velocity vector analysis indicates that the far-field airflow at the front of the vehicle is stable and high-speed (Figure 12), the front-end design of the NEV has a good deflection effect. The pressure cloud map of the vehicle's surface shows that a high-pressure area forms at the front of the vehicle, and the overall pressure distribution is uniform, effectively reducing additional resistance and improving stability and fuel economy (Figure 13). In the residual analysis of the wind resistance coefficient, the residuals converge with each iteration, and the final c_d value stabilises at 0.0008, which is much lower than the industry standard of 0.001, verifying the calculation accuracy (Figure 14). The velocity vector diagram at the mid-section shows a reasonable change in velocity gradient, and the airflow at the rear of the vehicle is stable (Figure 15). The pressure cloud map at the mid-section shows that the high pressure at the front of the vehicle, the smooth transition of the side pressure, and the low-pressure area at the rear all conform to theoretical expectations, indicating that the overall design is conducive to reducing wind resistance (Figure 16). The 3D curve is smooth, with a curvature change rate less than 0.05, confirming the qualification of the model calculation indicators (Figure 17).

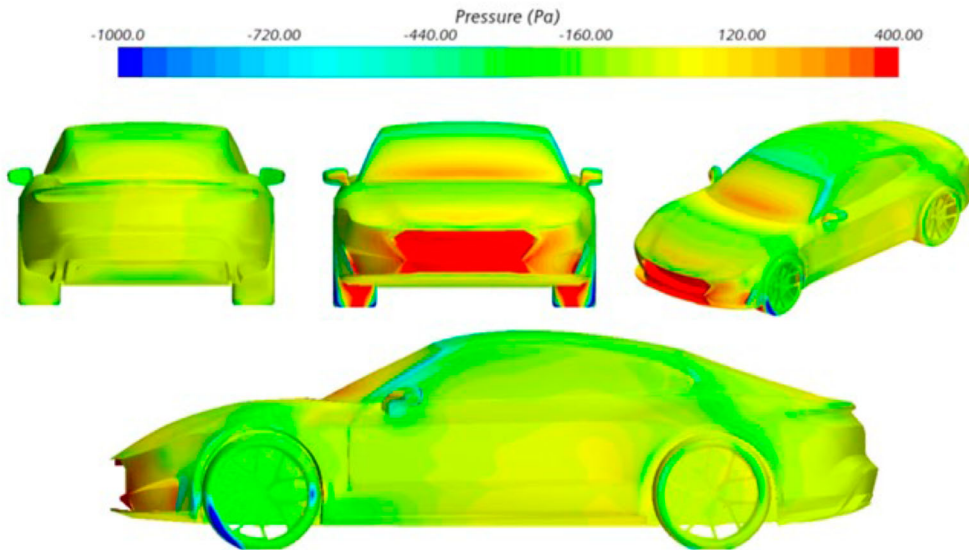


Figure 13. Pressure contour map of the vehicle surface.

By synthesising the evaluation results of multiple standards and verifying them through CFD, this study generates a NEV positive face shape design scheme that best meets the user's emotional intention and has good data in various performances, effectively proposing a novel design method. In addition, it provides a new perspective on sustainable, innovative design for enterprises and designers, thus confirming the validity and practical significance of the proposed research methods.

4.8. Kansei evaluation of the front view of the car

15 experts in the field of automotive design were selected as participants in this study. For three typical front designs of cars, a seven-point Likert scale (1 = highly mismatched, 7 = highly matched) was used to evaluate the matching degree of the generated three car front designs with the three sensory words 'exquisite', 'dynamic' and 'luxurious'. All participants had more than 10 years of experience in automotive design or related academic research, ensuring the professionalism and authority of the evaluation. The average scores of the three car front designs were 6.1, 6.56, and 6.25 (Table 9), all falling within the high-score range (6–7 points as the 'highly matched' interval), indicating that the experts had a high evaluation of the matching degree of car front designs with the Kansei words 'exquisite', 'dynamic' and 'luxurious', which means that the sensory characteristics of the car front designs are highly consistent with the experts' sensory cognition, and the emotional transmission effect of the design is significant.

To verify the validity of the research results, the questionnaire survey results were imported into SPSS for reliability and validity tests. The Cronbach's α coefficient was 0.81, which was higher than the standard threshold of 0.7, indicating that the seven-level Likert scale filled out by experts has good internal consistency reliability. This means that the three sensory words have stability in measuring the sensory intention of the car frontal design,

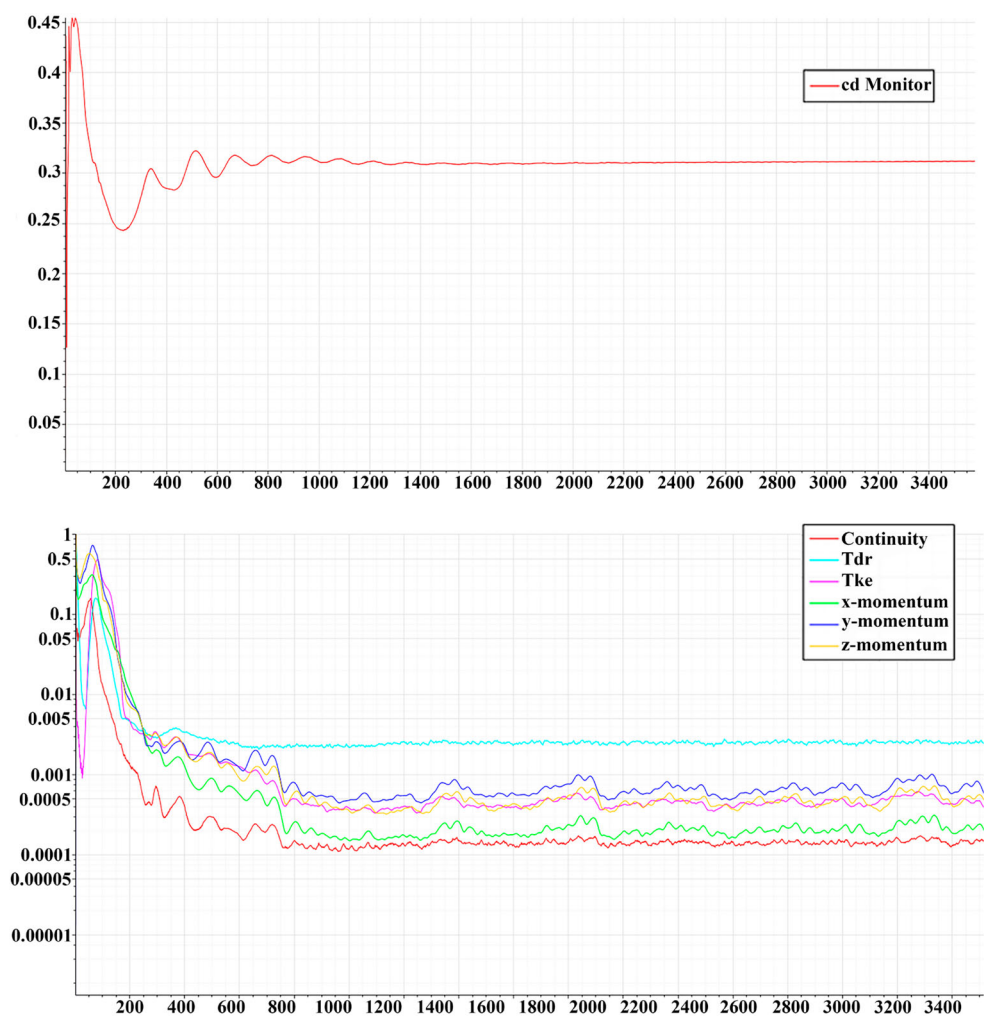


Figure 14. Wind resistance coefficient result.

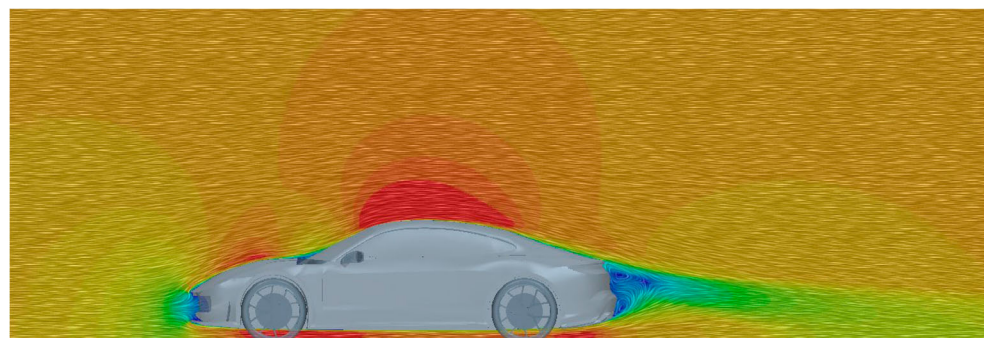


Figure 15. Mid-section velocity vector diagram.

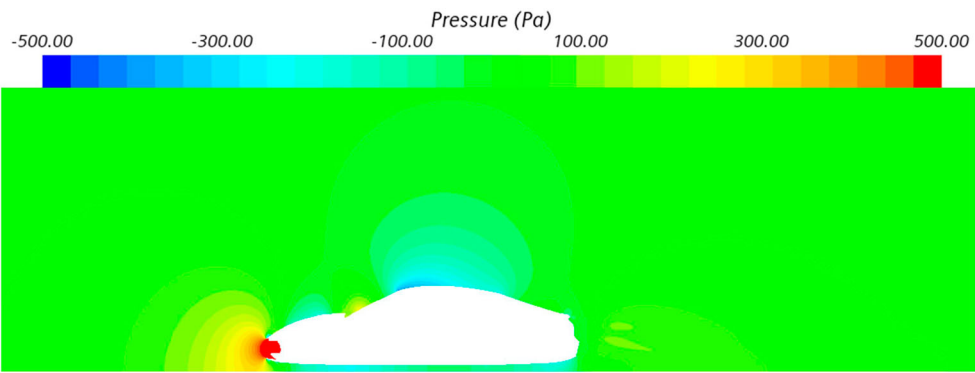


Figure 16. Mid-section pressure cloud image.

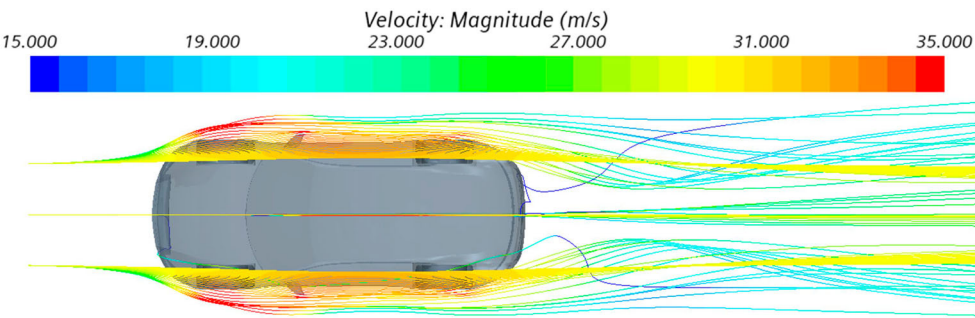


Figure 17. 3D curve.

Table 9. Expert subjective evaluation indicators.

	Exquisite	Dynamic	Luxurious
Average score	6.1	6.56	6.25
Cronbach α		0.81	
KMO		0.63	
Bartlett		20.25	

and the reliability of the expert evaluation results is relatively high. The scale was further tested using the Kaiser-Meyer-Olkin (KMO) value and the Bartlett's sphericity test results to jointly verify the structural validity of the scale. The results showed that the scale could effectively measure the matching degree between the car frontal design and the three key sensory words, and the research results were scientific and effective.

5. Result analysis and discussion

5.1. Analysis of the results of the product form feature combination

Entropy-AHP and SSA-XGBoost models are combined to obtain the final automotive product. First, Entropy-AHP accurately orders the Kansei needs of the user. Second, the SSA-XGBoost model accurately translates the Kansei needs into the design features of the

automotive product. Finally, the product is designed to meet the Kansei needs of the consumer effectively.

Based on the prediction results of SSA-XGBoost, a combination of design features corresponding to three key Kansei words is obtained. This result further demonstrates that in the current market environment, there is a strong and apparent demand for these three Kansei intentions in cars. 'Luxurious' Kansei intention reflects the user's pursuit of a high-quality driving experience, high-end interior materials and superior craftsmanship. The 'Dynamic' Kansei intention demonstrates the user's preference for dynamic performance, fun handling and sporty exterior design. The 'Exquisite' Kansei intention indicates that users expect the car to reach an extremely high level of detail, overall texture and design sophistication.

The SSA-XGBoost mapping model learns from the high-scoring design forms of 'Luxurious' in customer evaluations and predicts a brand-new design form for luxury cars. Among all the features of the front-face shape design, it exudes a luxurious atmosphere. The unique proportions and gorgeous metal trim strips further enhance the luxurious ambiance, allowing customers to feel a sense of dignity and extraordinariness, thus meeting users' Kansei demand for 'Luxurious' in cars. The prediction results of the 'Dynamic' model are also remarkable. The car as a whole adopts a highly-tensile streamlined design. The large-sized wheels and the exaggerated shape of the air intake not only enhance the vehicle's sense of sportiness but also meet customers' demand for 'sportiness' in vehicle design. In terms of 'Exquisite', the car guided by the model prediction has achieved an unprecedented level in detail processing. The body lines are finely polished, smooth and delicate, presenting an elegant texture under the light and shadow. The corners transition smoothly and roundly, eliminating any sense of abruptness, demonstrating an ultimate level of craftsmanship and adding a sense of refinement to the body.

Practice has proved that the combination of design features predicted by the SSA-XGBoost model perfectly matches the customer's Kansei demand for refined, luxurious and sporty cars, providing strong support for creating highly competitive automotive products in the market, which is expected to stand out in the fierce market competition and win the favour of customers.

5.2. Comparison of mapping accuracy between SSA-XGBoost and other neural networks

To verify the superiority of the proposed SSA-XGBoost model, models such as HO-XGBoost, TCN-Transformer-GRU and CNN were imported to conduct Kansei classification and prediction in this stage. To evaluate their generalisation ability and robustness. Firstly, the performance, accuracy and error of the 7 models on the Kansei dataset were observed separately. The R^2 and RMSE data of the 7 prediction models (Table 10) were compared. The results showed that the R^2 and RMSE of SSA-XGBoost in both the training set and the test set demonstrated better performance. These models (Figure 18) were trained and tested based on the shape combinations and affective evaluation values of 99 car frontal shapes. It was found that the prediction accuracy of SSA-XGBoost in the test set and the training set was both higher than 0.95, with an RMSE of 0.04 (Figures 19 and 20). In terms of predicting the mapping relationship of customer emotions, it demonstrated higher prediction accuracy and generalisation ability.

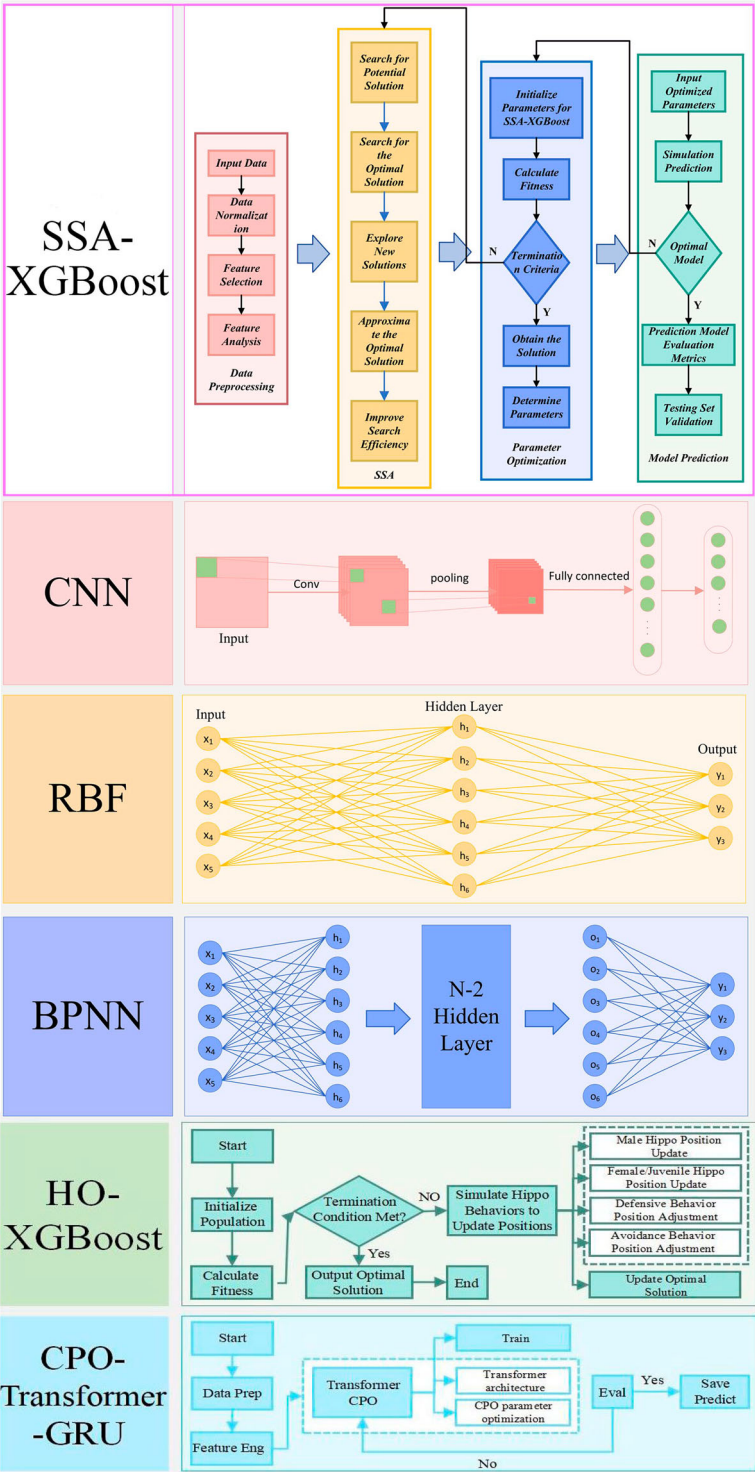


Figure 18. Comparison of SSA XGBoost architecture with other neural networks.

Table 10. Comparison of mapping accuracy between SSA-XGBoost and other neural networks.

SSA-XGBoost				CNN				RBF				BPNN				GRU			
R^2		RMSE		R^2		RMSE		R^2		RMSE		R^2		RMSE		R^2		RMSE	
Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
0.96	0.95	0.04	0.04	0.55	0.70	0.10	0.13	0.91	0.90	0.06	0.06	0.87	0.90	0.05	0.05	0.92	0.94	0.05	0.05
HO-XGBoost								CPO-Transformer-GRU											
R^2				RMSE				R^2						RMSE					
Train		Test		Train		Test		Train		Test		Train		Test		Train		Test	
0.91		0.93		0.44		0.37		0.21		0.29		0.23		0.21					

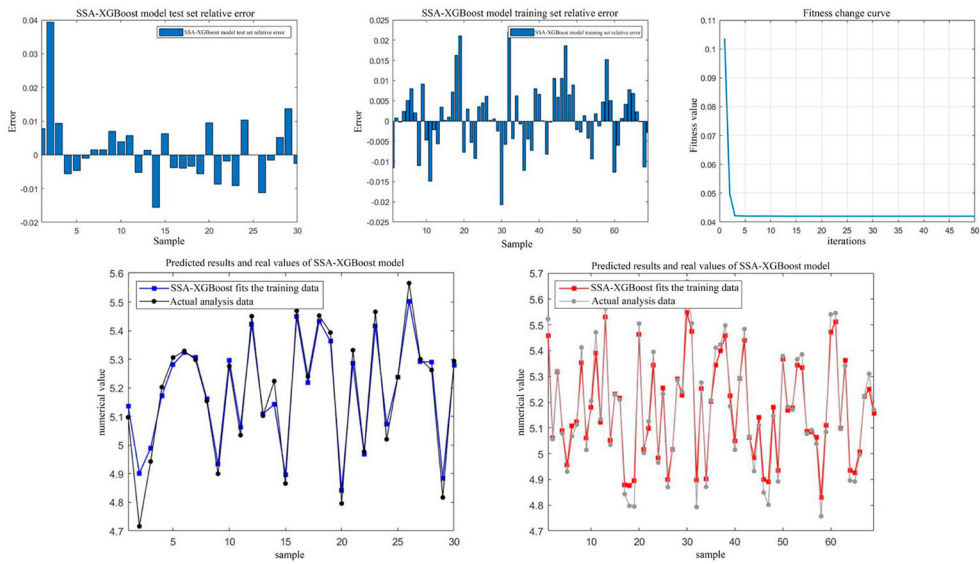


Figure 19. Predicted and actual values generated by SSA-XGBoost.

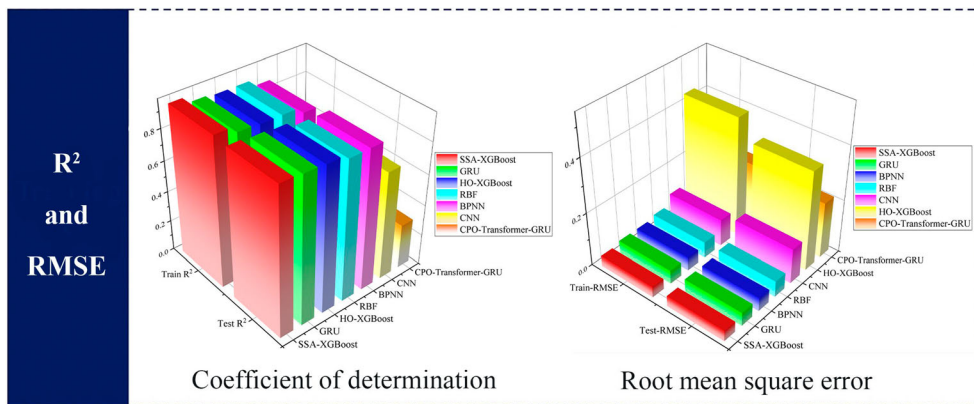


Figure 20. Comparison of SSA-XGBoost with other neural networks.

6. Conclusion

This paper combines KE and SSA-XGBoost to achieve the front-face shape design of NEV. First, 12 key Kansei adjectives and 99 front-face images of NEV are generated and extracted through large-language models and SDM. Then, the Entropy-AHP method is used to rank the 12 Kansei words. After that, the morphological analysis table is deconstructed and redrawn by the morphological analysis method. The SSA-XGBoost algorithm constructs a mapping model between customer Kansei and NEV frontal product design features, predicting the optimal NEV frontal design combination that can meet the high emotional value of customers. Finally, the SDM and Rhino modelling software establish three-dimensional models based on two-dimensional images, and conduct CFD simulation experiments and expert perceptual evaluations to comprehensively consider the effectiveness of the design

scheme. The Kansei design method constructed in this paper can meet customers' complex Kansei needs and provide creative references for the development of new NEV products. The main innovations of this paper are summarised as follows:

- Traditional methods of obtaining Kansei words in KE are often time-consuming and labour-intensive. In contrast, the Kansei words obtained through ChatGPT can be acquired at a very high speed, providing more flexible data references for new product development.
- SDM can automatically generate corresponding 3D car renderings based on a single front view feature, significantly enhancing innovation and design efficiency.
- The SSA-XGBoost can effectively enhance the model's ability to capture and predict, enabling it to capture the complex relationships among features, thereby improving the understanding and prediction performance of Kansei data.

Although this study has obtained three front-face form design schemes for NEV, there are still some limitations that need to be addressed. First, the current research mainly focuses on the relationship between visual design and customers' emotions. In the future, other sensory factors (such as touch, sound, and materials) can be incorporated into the model. Through the fusion of multi-modal data, users' Kansei responses can be comprehensively captured, further enhancing the overall Kansei analysis ability for NEV product design. Second, due to the high production cost of NEV products and limited research funds, this study only uses three-dimensional models for simulation and prediction to analyse the feasibility of the design schemes. There may be deviations between the simulation results and the final mass-produced physical products. Finally, this paper mainly explores the front-face form design of NEV. In subsequent research, it is possible to explore further how to apply this design method to more aspects of automotive design, such as material selection, structural design, and energy-efficiency optimisation. For example, the design of automobile steering wheels, seats, and tires.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This study was funded by the National Natural Science Foundation of China [62206118], Jiangxi Province Degree and Graduate Education Teaching Reform Project [JXYJG-2024-008], Nanchang University Youth Talent Cultivation Innovation Fund Project [XX202506030027].

Data availability statement

Data will be made available on request.

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